Math 342W / 642 / RM742 Spring 2026
Midterm Examination One

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Full Name _____

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Cheating Using or attempting to use unauthorized assistance, material, or study aids in examinations or other academic work or preventing, or attempting to prevent, another from using authorized assistance, material, or study aids. Example: using an unauthorized cheat sheet in a quiz or exam, altering a graded exam and resubmitting it for a better grade, etc.

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Instructions

This exam is 110 minutes (variable time per question) and closed-book. You are allowed **two** pages (front and back) of a "cheat sheet", blank scrap paper (provided by the proctor) and a graphing calculator (which is not your smartphone). Please read the questions carefully. Within each problem, I recommend considering the questions that are easy first and then circling back to evaluate the harder ones. No food is allowed, only drinks.

Problem 1 George Box and Norman Draper in 1987 wrote "All models are wrong but some are useful". Below are some conceptual questions about this aphorism and modeling in general. Let $< \boldsymbol{X}, \boldsymbol{y} >$ be a historical dataset. Let $\boldsymbol{x}_*$ refer to a future object that is not a row in the historical dataset $\boldsymbol{X}$.

(a) [11 pt / 11 pts] For each true statement, circle the roman numeral on the left.

(i) TRUE In the quote, "models are wrong" since $\hat{y}_* := g(\boldsymbol{x}_*) \neq y_*$ for all $\boldsymbol{x}_*$.

(ii) TRUE In the quote, "models are wrong" since they are approximations to reality.

(iii) In the quote, "models are wrong" since you cannot get an honest assessment of how well they will predict on average for $\boldsymbol{x}_*$.

(iv) In the quote, "models are wrong" but in context they only were speaking about non-mathematical models.

(v) In the quote, "models are wrong" only in the case where $\mathcal{Y} \nsubseteq \mathbb{R}$.

(vi) TRUE In the quote, "some [models] are useful" since they are approximations to reality.

(vii) In the quote, "some [models] are useful" only because you can compute oos error metrics.

(viii) In the quote, "some [models] are useful" since g_{final} is more accurate than g if you do a training-test split

(ix) In the quote, "some [models] are useful" but in context they meant only those that are mathematical.

(x) TRUE In the quote, "some [models] are useful" because they beat g_0.

(xi) In the quote, "some [models] are useful" because they give you the ability to compute $\hat{y}_* := g(\boldsymbol{x}_*)$.

Problem 2 Let $< \boldsymbol{X}, \boldsymbol{y} >$ be a historical dataset with $n = 300$. Assume the first column of $\boldsymbol{X}$ is $\mathbf{1}_n$ so that there are $p + 1$ columns. Let the sets A, B, C denote random subsets of the indicies such that $A \cup B \cup C = \{1, \ldots, n\}$ but none of the sets have any overlap, i.e., $A \cap B = A \cap C = B \cap C = \varnothing$ and each set is equally sized. Let $\text{nrow}(U)$, $\text{ncol}(U)$ denote the number of rows and columns in U respectively. Let $U = \boldsymbol{QR}$ be the Q-R decompositions of the the matrix on the left hand side. Let the subscript of X and $\boldsymbol{y}$ denote only those rows, e.g., $\boldsymbol{X}_A$ is the submatrix $\boldsymbol{X}$ with only the rows of A. Also let $\boldsymbol{Q}_S$, $\boldsymbol{R}_S$ denote the Q-R decomposition of the matrix $\boldsymbol{X}_S$ where S is a subset of $\{1, \ldots, n\}$. Assume $\boldsymbol{X}_A, \boldsymbol{X}_B, \boldsymbol{X}_C$ are full rank.

(a) [18 pt / 29 pts] For each true statement, circle the roman numeral on the left.

(i) TRUE $\text{nrow}(\boldsymbol{X}_A) + \text{nrow}(\boldsymbol{X}_B) + \text{nrow}(\boldsymbol{X}_C) = n$

(ii) $\text{ncol}(\boldsymbol{X}_A) + \text{ncol}(\boldsymbol{X}_B) + \text{ncol}(\boldsymbol{X}_C) = p + 1$

(iii) $\text{colsp}[\boldsymbol{X}_A] = \text{colsp}[\boldsymbol{X}_B] = \text{colsp}[\boldsymbol{X}_C]$

(iv) TRUE $\text{rank}[\boldsymbol{X}_A] = \text{rank}[\boldsymbol{X}_B] = \text{rank}[\boldsymbol{X}_C]$

(v) TRUE $\text{nrow}(\boldsymbol{Q}_A) + \text{nrow}(\boldsymbol{Q}_B) + \text{nrow}(\boldsymbol{Q}_C) = n$

(vi) $\text{ncol}(\boldsymbol{Q}_A) + \text{ncol}(\boldsymbol{Q}_B) + \text{ncol}(\boldsymbol{Q}_C) = p + 1$

(vii) $\text{colsp}[\boldsymbol{Q}_A] = \text{colsp}[\boldsymbol{Q}_B] = \text{colsp}[\boldsymbol{Q}_C]$

(viii) TRUE $\text{rank}[\boldsymbol{Q}_A] = \text{rank}[\boldsymbol{Q}_B] = \text{rank}[\boldsymbol{Q}_C]$

(ix) TRUE $\text{rank}[\boldsymbol{R}_A] = \text{rank}[\boldsymbol{R}_B] = \text{rank}[\boldsymbol{R}_C]$

(x) TRUE The matrix $\boldsymbol{QR}$ is full rank

(xi) TRUE The matrix $\boldsymbol{XR}$ is full rank

(xii) TRUE $\boldsymbol{Q}_A^\top \boldsymbol{Q}_A = \boldsymbol{I}_{p+1}$

(xiii) $\boldsymbol{Q}_A^\top \boldsymbol{Q}_B = \boldsymbol{I}_{p+1}$

(xiv) $\boldsymbol{Q}_A^\top \boldsymbol{Q}_C = \boldsymbol{I}_{p+1}$

(xv) $\boldsymbol{Q}_A \boldsymbol{Q}_A^\top = \boldsymbol{I}_{n/3}$

(xvi) $\boldsymbol{Q}_A \boldsymbol{Q}_B^\top = \boldsymbol{I}_{n/3}$

(xvii) $\boldsymbol{Q}_A \boldsymbol{Q}_C^\top = \boldsymbol{I}_{n/3}$

(xviii) $\boldsymbol{Q}_A \boldsymbol{Q}_A^\top \boldsymbol{y}_A = \boldsymbol{Q}_B \boldsymbol{Q}_B^\top \boldsymbol{y}_B = \boldsymbol{Q}_C \boldsymbol{Q}_C^\top \boldsymbol{y}_C$

Let $\mathcal{A}$ be an algorithm and $\mathcal{H}$ be a candidate set. Let $g = \mathcal{A}(< \boldsymbol{X}, \boldsymbol{y} >, \mathcal{H})$ and $g_S = \mathcal{A}(< \boldsymbol{X}_S, \boldsymbol{y}_S >, \mathcal{H})$ where S is a subset of $\{1, \ldots, n\}$. The word "expected" below means on average over many possible datasets.

(b) [3 pt / 32 pts] Which of the following models have the lowest in-sample error? Circle the roman numeral on the left. If they are all equal, or the answer cannot be determined given the information available, do not circle any.

(i) g_A

(ii) $g_{A \cup B}$

(iii) g

(c) [3 pt / 35 pts] Which of the following models have the lowest **expected** in-sample error? Circle the roman numeral on the left. If they are all equal, or the answer cannot be determined given the information available, do not circle any.

(i) g_A

(ii) $g_{A \cup B}$

(iii) g

(d) [3 pt / 38 pts] Which of the following models have the lowest **expected** oos error? Circle the roman numeral on the left. If they are all equal, or the answer cannot be determined given the information available, do not circle any.

(i) g_A

(ii) $g_{A \cup B}$

(iii) g

(e) [3 pt / 41 pts] Which of the following models have the lowest **expected** estimation error? Circle the roman numeral on the left. If they are all equal, or the answer cannot be determined given the information available, do not circle any.

(i) g_A

(ii) $g_{A \cup B}$

(iii) g

(f) [3 pt / 44 pts] Which of the following models have the lowest **expected** misspecification error? Circle the roman numeral on the left. If they are all equal, or the answer cannot be determined given the information available, do not circle any.

(i) g_A

(ii) $g_{A \cup B}$

(iii) g

(g) [3 pt / 47 pts] Which of the following models have the lowest **expected** ignorance error? Circle the roman numeral on the left. If they are all equal, or the answer cannot be determined given the information available, do not circle any.

(i) g_A

(ii) $g_{A \cup B}$

(iii) g

(h) [3 pt / 50 pts] Compare the expected error of $g_A(\boldsymbol{X}_A)$ and $g_A(\boldsymbol{X}_B)$. In the circle below, write $<, >$ or $=$.

$$\text{Error}(g_A(\boldsymbol{X}_A)) \quad < \quad \text{Error}(g_A(\boldsymbol{X}_B))$$

(i) [13 pt / 63 pts] Circle the roman numeral on the left for all the error measurements that are *honest* i.e., they represent the generalization error in future predictions without further assumptions.

(i) $\text{Error}(g_A(\boldsymbol{X}_A))$

(ii) honest $\text{Error}(g_A(\boldsymbol{X}_B))$

(iii) honest $\text{Error}(g_A(\boldsymbol{X}_C))$

(iv) honest $\text{Error}(g_A(\boldsymbol{X}_{B \cup C}))$

(v) $\text{Error}(g_A(\boldsymbol{X}))$

(vi) $\text{Error}(g_{A\cup B}(\boldsymbol{X}_A))$

(vii) $\text{Error}(g_{A\cup B}(\boldsymbol{X}_B))$

(viii) honest $\text{Error}(g_{A\cup B}(\boldsymbol{X}_C))$

(ix) $\text{Error}(g_{A\cup B}(\boldsymbol{X}_{B\cup C}))$

(x) $\text{Error}(g_{A\cup B}(\boldsymbol{X}))$

(xi) $\text{Error}(g(\boldsymbol{X}_A))$

(xii) $\text{Error}(g(\boldsymbol{X}_B))$

(xiii) $\text{Error}(g(\boldsymbol{X}_C))$

Assume SSE is the error metric going forward.

(j) [3 pt / 66 pts] Write a formula to find the in-sample error of the model g_A. Be as explicit as possible using the notation given in this problem.

$$(\boldsymbol{y}_A - g_A(\boldsymbol{X}_A))^\top(\boldsymbol{y}_A - g_A(\boldsymbol{X}_A)) = \sum_{i\in A}(y_i - g_A(\boldsymbol{x}_{i\cdot}))^2$$

The sum notation can be used for the next three questions as well.

(k) [3 pt / 69 pts] Write a formula to find the *most accurate* measure of the oos error of the model g_A. Be as explicit as possible using the notation given in this problem.

$(\boldsymbol{y}_{B\cup C} - g_A(\boldsymbol{X}_{B\cup C}))^\top(\boldsymbol{y}_{B\cup C} - g_A(\boldsymbol{X}_{B\cup C}))$

(l) [3 pt / 72 pts] Now assume n is very large, i.e. $n >> 300$. Write a formula to find the *most accurate* measure of the oos error of the model g_A. Be as explicit as possible using the notation given in this problem.

$(\boldsymbol{y} - g_A(\boldsymbol{X}))^\top(\boldsymbol{y} - g_A(\boldsymbol{X}))$

(m) [3 pt / 75 pts] Write a formula to find an estimate of the oos error of the model g using $K = 3$-fold cross validation. Be as explicit as possible using the notation given in this problem.

$$\begin{aligned}&(\boldsymbol{y}_C - g_{A\cup B}(\boldsymbol{X}_C))^\top(\boldsymbol{y}_C - g_{A\cup B}(\boldsymbol{X}_C)) + \\ &(\boldsymbol{y}_B - g_{A\cup C}(\boldsymbol{X}_B))^\top(\boldsymbol{y}_B - g_{A\cup C}(\boldsymbol{X}_B)) + \\ &(\boldsymbol{y}_A - g_{B\cup C}(\boldsymbol{X}_A))^\top(\boldsymbol{y}_A - g_{B\cup C}(\boldsymbol{X}_A))\end{aligned}$$

(n) [3 pt / 78 pts] Will the estimate in the previous question be conservative? Yes/no and why.

Yes. The final model g will have lower estimation error for future predictions than the estimate found in the previous question.

Problem 3

Consider the following data frame `D` illustrated using the `skimr::skim` function:

```
-- Data Summary ------------------------
                           Values
Name                       D
Number of rows             137
Number of columns          8
_______________________
Column type frequency:
  factor                   2
  numeric                  6
________________________
Group variables            None

-- Variable type: factor ----------------------------------------------
  skim_variable n_missing complete_rate ordered n_unique top_counts
1 trt                   0             1 FALSE          2 con: 69, exp: 68
2 celltype              0             1 FALSE          4 sma: 48, squ: 35, ade: 27, lar: 27

-- Variable type: numeric ---------------------------------------------
  skim_variable n_missing complete_rate    mean      sd p0 p25 p50 p75 p100
1 time                  0             1 122.    158.     1  25  80 144  999
2 status                0             1   0.934   0.249  0   1   1   1    1
3 karno                 0             1  58.6    20.0   10  40  60  75   99
4 diagtime              0             1   8.77   10.6    1   3   5  11   87
5 age                   0             1  58.3    10.5   34  51  62  66   81
6 prior                 0             1   2.92    4.56   0   0   0  10   10
```

Let $y :=$ `time`. And let $\boldsymbol{X}$ denote the design matrix constructed from the remaining columns of `D`. Assume the first column of $\boldsymbol{X}$ is $\mathbf{1}_n$.

(a) [2 pt / 80 pts] What type of modeling task is this? In this class, the answer is a one-word answer.

regression

(b) [2 pt / 82 pts] What is the data type of `celltype`?

unordered / nominal categorical / factor

(c) [3 pt / 85 pts] What is the *most likely* data type of `status`? Be as specific as possible.

binary / dummy / categorical with two levels: 0,1

(d) [4 pt / 89 pts] What is the value of p?

(4-1) for the dummies in `celltype`
+ (2-1) for the dummy in `trt`
+ 5 for the remaining numeric features

= 9

(e) [4 pt / 93 pts] What is g_0?

122

(f) [5 pt / 98 pts] How do we know g_0 will perform poorly in-sample for this dataset?

$\text{sd}(y) = 158$ and the range of $y \in (0, 999)$ which are both large.

(g) [3 pt / 101 pts] Compute R^2 for g_0.

You don't need to compute anything because by definition it is 0.

(h) [4 pt / 105 pts] Without any other information given, do you think OLS will beat g_0 oos? Yes, no or can't be determined. Explain.

Can't be determined. We have no idea how informative these 7 features are without understanding context of the phenomenon we are trying to model.

(i) [3 pt / 108 pts] We run an OLS on all features and calculate SSE = 2482093. Compute RMSE to once decimal.

$$RMSE = \sqrt{\frac{SSE}{n-(p+1)}} = \sqrt{\frac{2482093}{137-(9+1)}} = 139.8$$

(j) [5 pt / 113 pts] We run an OLS on all features and calculate SSE = 2482093. Compute R^2 to three decimals.

$$R^2 := 1 - \frac{SSE}{SST} = 1 - \frac{SSE}{(n-1)s_y^2} = 1 - \frac{2482093}{(137-1)158^2} = .267$$

(k) [2 pt / 115 pts] Assume there was no overfitting. What is the *least likely* source of error in the OLS model?

estimation

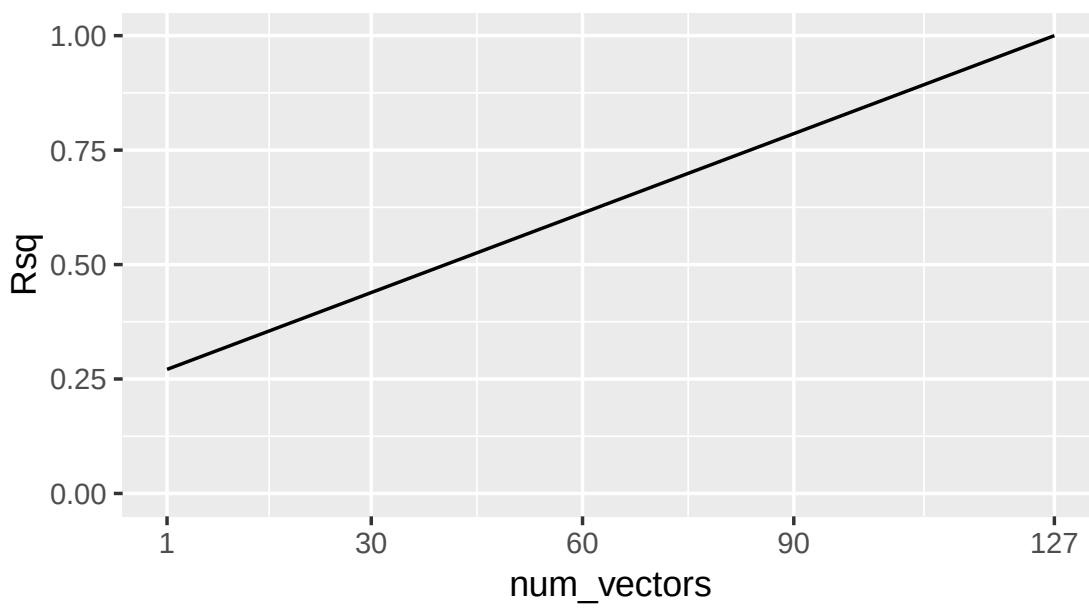
Here are the first six rows:

```
> head(D)
             trt  celltype time status karno diagtime age prior
107 experimental smallcell   29      1    40        8  67     0
121 experimental     adeno  186      1    90        3  60     0
51       control     adeno   12      1    50        4  63    10
58       control     large  553      1    70        2  47     0
15       control  squamous   11      1    70       11  48    10
81  experimental  squamous   25      1    20       36  63     0
```

(l) [6 pt / 121 pts] Provide the first six rows of $\boldsymbol{X}$. No need to name the columns or rows.

(Intercept)	trt_exp	cell_large	cell_small	cell_squam	status	karno	diagtime	age	prior
1	1	0	1	0	1	40	8	67	0
1	1	0	0	0	1	90	3	60	0
1	0	0	0	0	1	50	4	63	10
1	0	1	0	0	1	70	2	47	0
1	0	0	0	1	1	70	11	48	10
1	1	0	0	1	1	20	36	63	0

(m) [8 pt / 129 pts] Consider the scenario where you add junk vectors (full of random values) one-by-one and run OLS for each new junk vector. You keep adding junk vectors until OLS can no longer compute. Assume each iteratively new $\boldsymbol{X}$ is full rank. Graph the expected R^2 from an OLS on the y axis and the number of vectors *in addition* to the features already in $\boldsymbol{X}$ on the x-axis. The y-axis should start at zero and end at 1. Label axes.



Now assume $y :=$ `trt`

(n) [2 pt / 131 pts] What type of modeling task is this? In this class, the answer is a one- or two-word answer.

binary classification

(o) [4 pt / 135 pts] Explain two reasons why the perceptron will likely not be a good model here.

(1) It is doubtful the data is linearly separable and (2) the linear hyperplane threshold model will likely suffer from misspecification error.

Now assume you use KNN to model the response.

(p) [4 pt / 139 pts] Provide an explicit distance function $d(\boldsymbol{x}_*, \boldsymbol{x}_i)$ where $\boldsymbol{x}_i$ is a row in $\boldsymbol{X}$ and $\boldsymbol{x}_*$ is a future subject we wish to predict their response for. Be as explicit as possible in your notation

$$\text{e.g., the Euclidean distance: } d(\boldsymbol{x}_*, \boldsymbol{x}_i) = ||\boldsymbol{x}_* - \boldsymbol{x}_i||^2 := \sum_{j=1}^{9} (x_{i,j} - x_{*,j})^2$$

Regardless of what you wrote in the previous question, assume the distance function is reasonable and standard going forward.

(q) [3 pt / 142 pts] Explain why KNN will likely beat the SVM with the Vapnik objective function.

The KNN will likely have less misspecification error than the SVM with the Vapnik objective function (which is a linear model).

(r) [4 pt / 146 pts] If $K = n$, what is the prediction for the third row in `D`?

$K = n$ implies there is one large cluster of all data points. This means g_0 which is the mode. The mode of `trt` is `control`.

(s) [4 pt / 150 pts] If $K = 1$, what is the prediction for the third row in `D`?

$K = 1$ implies you look at the closest neighbor which (if you're predicting in-sample) is yourself. This means the prediction is the same as the actual response of the third row which is `control`.